

A Lean-Verified 51/50 Cutoff for Four-Arc Minimizers

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Package status: VERIFIED AND SEMANTICALLY ACCEPTED

A modeled four-arc exclusion for the strip density

Statement and meaning of minimization

Fix a real number $\lambda \geq 51/50$, and put

$$f_\lambda(x, y) = \begin{cases} 1, & |y| \leq 1, \\ \lambda, & |y| > 1. \end{cases}$$

For a measurable planar carrier E , write $A_\lambda(E) = \int_E f_\lambda d\mathcal{L}^2$. In this package the perimeter of a competitor is **canonically** $P_\lambda(E) = \int_{\partial E} f_\lambda d\mathcal{H}^1$, where ∂E is its complete topological frontier and $\mathcal{H}^1$ is Euclidean one-dimensional Hausdorff measure. The density on both interfaces is one. These choices are part of the statement, not conventions left to the competitor. The coordinate plane is transported to its Euclidean L^2 realization before Hausdorff measure is formed. [StripDensity \(audit\)](#) [WeightedArea \(audit\)](#) [FrontierMeasure \(audit\)](#) [WeightedPerimeter \(audit\)](#) [planeEuclideanHomeomorph \(audit\)](#)

The admissible class contains the empty carrier, general measurable carriers with integrable weighted area and integrable weighted complete-frontier measure, and the explicit four-arc and cap-replacement constructors. In particular it is not a comparison class consisting only of four-arc curves. The name finite perimeter in the package refers to these recorded integrability conditions; it is not an identification with all Caccioppoli sets modulo null sets. [FinitePerimeterRegion \(audit\)](#) [AdmissibleCompetitor \(audit\)](#) [AdmissibleCompetitor.weightedArea_eq_carrier \(audit\)](#) [AdmissibleCompetitor.WeightedPerimeter \(audit\)](#)

Theorem. Let C be the modeled four-arc carrier with curvature parameter h and exterior half-angle α , satisfying

$$0 < h \leq 1, \quad 0 < \alpha < \pi/2, \quad 1 < \lambda, \quad \lambda \cos \alpha = h.$$

If $\lambda \geq 51/50$, then C is not a weighted-perimeter minimizer in the preceding admissible class. Thus there is an admissible competitor with the same weighted area and strictly smaller weighted perimeter. The constructor includes $h = 1$; this endpoint is not silently removed by the word regular. The formal CMV hypotheses are precisely the density jump and incidence equation displayed above, in addition to the constructor inequalities. They do not assume minimality, a certificate, or a perimeter comparison.

[FourArcCandidate \(audit\)](#) [FourArcCandidate.SatisfiesCMVTypeIVHypotheses \(audit\)](#)

[FourArcCandidate.IsWeightedPerimeterMinimizer \(audit\)](#)

[CMVPublishedCutoff51_50.candidate_not_isWeightedPerimeterMinimizer_from_51_50 \(audit\)](#)

The source motivation is Cañete, Miranda Jr., and Vittone, *Some Isoperimetric Problems in Planes with Density*, §3.2: Lemma 3.8 lists the strip candidates, equations (26)–(27) give the type-(iii) and type-(iv) formulas, Conjecture 3.12 describes the expected isoperimetric picture, and Theorem 3.16 with Remark 3.17 supplies the cap-replacement comparison and discusses its limitation. The stable source is arXiv:0906.1256v1, especially the printed manuscript pages 19 and 22–24; the journal publication is *Journal of Geometric Analysis* **20** (2010), 243–290. The theorem here concerns the explicit model just defined, not an unformalized classification theorem imported from that article.

1. Exact geometry and scalar formulas

The four-arc carrier consists of a central strip core and congruent upper and lower exterior circular caps. Set $R = 1/h$ and $c = 2\sin\alpha/h$. Each exterior cap has chord c and area $(\alpha - \sin\alpha\cos\alpha)/h^2$; the side arcs of the core have half-angle $\arcsin h$. The incidence equation and the recorded principal-angle range give $\alpha = \arccos(h/\lambda)$. [FourArcCandidate.capChord \(audit\)](#)

[FourArcCandidate.capArea](#) (audit) [FourArcCandidate.stripCore](#) (audit) [FourArcCandidate.assembly](#) (audit)
[FourArcCandidate.alpha_eq_arccos](#) (audit)

All area and perimeter identities used below are identities for these carriers, rather than definitions of an arbitrary scalar objective. Circular-segment areas are integrated, the full frontier is identified, the Euclidean Hausdorff length of each arc is computed, and the finitely many junctions have zero $\mathcal{H}^1$ measure. Consequently no chord glued inside the carrier is charged as exposed boundary.

[FourArcAssembly.weightedArea_formula](#) (audit) [FourArcAssembly.frontier_carrier](#) (audit)
[fourArc_frontier_weightedPerimeter_eq](#) (audit) [FourArcCandidate.candidate_area_formula](#) (audit)
[FourArcCandidate.candidate_perimeter_formula](#) (audit)

For fixed $\lambda > 1$, introduce the following notation. For type (iii), let $q = 2s - 1$, with $0 < s < 1$, and set

$$B_3(s) = \lambda \arccos(q/\lambda) + \arcsin q + \pi/2, \quad D_3(s) = \frac{\sqrt{\lambda^2 - q^2}}{\lambda} - \sqrt{1 - q^2}.$$

For type (iv), with $0 < t \leq 1$, set

$$B_4(t) = \lambda \arccos(t/\lambda) + \arcsin t, \quad D_4(t) = \frac{\sqrt{\lambda^2 - t^2}}{\lambda} - \sqrt{1 - t^2}.$$

The exact formulas are

$$A_3(s) = \frac{B_3(s) + (2s + 1)D_3(s)}{s^2}, \quad P_3(s) = \frac{2(B_3(s) + D_3(s))}{s},$$

$$A_4(t) = \frac{2(B_4(t) + tD_4(t))}{t^2}, \quad P_4(t) = \frac{4B_4(t)}{t}.$$

[LeanSuffixAnalytic.typeThreeShape](#) (audit) [LeanSuffixAnalytic.typeThreeAngle](#) (audit)
[LeanSuffixAnalytic.typeThreeDelta](#) (audit) [LeanSuffixAnalytic.typeThreeArea](#) (audit)
[LeanSuffixAnalytic.typeThreePerimeter](#) (audit) [LeanSuffixAnalytic.typeFourAngle](#) (audit)
[LeanSuffixAnalytic.typeFourDelta](#) (audit) [LeanSuffixAnalytic.typeFourArea](#) (audit)
[LeanSuffixAnalytic.typeFourPerimeter](#) (audit)

For every $s \in (0, 1)$ the type-(iii) constructor realizes $A_3(s), P_3(s)$ by a genuine carrier: a strip core, one upper exterior cap, two side arcs, and the exposed bottom segment. It allows the exterior major-cap branch when $s < 1/2$, and the zero-length bottom segment at $s = 1/2$. Those branches matter because an equal-area root need not stay above $1/2$. Its full frontier, not a hand-picked list omitting the bottom segment, has the stated perimeter. Integrability makes this carrier an admissible general competitor. [TypeThreeAssembly](#) (audit) [TypeThreeAssembly.frontier_carrier](#) (audit) [TypeThreeAssembly.weightedArea_formula](#) (audit) [TypeThreeAssembly.weightedPerimeter_formula](#) (audit) [TypeThreeAssembly.hasModeledCompetitor](#) (audit) [CMVSuffixModel.TypeThreeAssembly.scalarWeightedArea_eq_typeThreeArea](#) (audit) [CMVSuffixModel.TypeThreeAssembly.scalarWeightedPerimeter_eq_typeThreePerimeter](#) (audit)

The four-arc carrier satisfies $A_\lambda(C) = A_4(h)$ and $P_\lambda(C) = P_4(h)$, including $h = 1$. Therefore it suffices to produce $s \in (0, 1)$ with $A_3(s) = A_4(h)$ and $P_3(s) < P_4(h)$. The bridge to nonminimality uses the realized competitor just described.

[CMVSuffixModel.FourArcCandidate.weightedArea_eq_typeFourArea](#) (audit)
[CMVSuffixModel.FourArcCandidate.weightedPerimeter_eq_typeFourPerimeter](#) (audit)
[CMVSuffixModel.FourArcCandidate.not_isWeightedPerimeterMinimizer_of_analytic_typeThree](#) (audit)

2. One stationary comparison controls every curvature

Define the fold numerators $K_i(u) = u^3 A_i(u)$ on $0 < u < 1$. Explicitly, with $q = 2s - 1$,

$$K_3(s) = 4s\left(\frac{1+q}{\sqrt{1-q^2}} - \frac{\lambda^2+q}{\lambda\sqrt{\lambda^2-q^2}}\right) - 2(B_3(s) + D_3(s)),$$

$$K_4(t) = 4\left[t\left(\frac{1}{\sqrt{1-t^2}} - \frac{\lambda}{\sqrt{\lambda^2-t^2}}\right) - B_4(t)\right].$$

Differentiation also gives $P_i(u) = uA_i(u) = K_i(u)/u^2$. [LeanSuffixAnalytic.typeThreeFold](#) (audit)
[LeanSuffixAnalytic.typeFourFold](#) (audit) [LeanSuffixAnalytic.hasDerivAt_typeThreeArea](#) (audit)
[LeanSuffixAnalytic.hasDerivAt_typeFourArea](#) (audit)
[LeanSuffixAnalytic.hasDerivAt_typeThreePerimeter](#) (audit)
[LeanSuffixAnalytic.hasDerivAt_typeFourPerimeter](#) (audit)

For $|z| < 1$,

$$\frac{1}{(1-z^2)^{3/2}} - \frac{\lambda}{(\lambda^2 - z^2)^{3/2}} > 0.$$

Indeed $\lambda^2 - z^2 > \lambda^2 (1 - z^2)$ unless $z = 0$, and at zero the difference is $1 - \lambda^{-2} > 0$. The fold derivatives are respectively $16s^2$ times this difference at $z = 2s - 1$ and $4t^2$ times it at $z = t$. Hence both folds are strictly increasing on $(0, 1)$. A zero t_* of K_4 is unique; A_4 decreases before it and increases after it, with $A_4(t_*) \leq A_4(t)$ also at $t = 1$.

[LeanSuffixAnalytic.hasDerivAt_typeThreeFold \(audit\)](#) [LeanSuffixAnalytic.hasDerivAt_typeFourFold \(audit\)](#) [LeanSuffixAnalytic.typeThreeFold_strictMonoOn \(audit\)](#) [LeanSuffixAnalytic.typeFourFold_strictMonoOn \(audit\)](#) [LeanSuffixAnalytic.typeFourArea_fold_le \(audit\)](#)

Suppose now that a certificate supplies numbers

$$\frac{1}{2} < s_* < 1, \quad 0 < t_* < 1, \quad A_3(s_*) = A_4(t_*), \quad K_4(t_*) = 0, \\ K_3(s_*) < 0, \quad P_3(s_*) < P_4(t_*).$$

The stationary-pair record supplies the first line; the last two inequalities are separate obligations, not hidden in the record.

[LeanSuffixAnalytic.StationaryEqualAreaPair \(audit\)](#)

Since K_3 is increasing and negative at s_* , A_3 is strictly decreasing on $(0, s_*]$. Moreover $A_3(s) \rightarrow +\infty$ as $s \downarrow 0$. For every $t \in (0, 1]$, the inequality $A_4(t) \geq A_4(t_*) = A_3(s_*)$ therefore gives a unique descending-branch root $r(t) \in (0, s_*]$ of $A_3(r(t)) = A_4(t)$, with $K_3(r(t)) < 0$. Strict increase of K_3 identifies this descending root globally, even if the full area equation has another root.

[LeanSuffixAnalytic.typeThreeArea_tendsto_atRight_zero \(audit\)](#) [LeanSuffixAnalytic.typeThreeArea_descending_root_existsUnique \(audit\)](#) [LeanSuffixAnalytic.typeThreeArea_descending_root_existsUnique_of_stationary_pair \(audit\)](#) [LeanSuffixAnalytic.DescendingEqualAreaEnvelope \(audit\)](#)

The exact same-curvature comparison $A_3(u) < A_4(u)$, valid for $\lambda > 1$ and $0 < u \leq 1$, implies the stronger root order $r(t) < t$ after the descending branch has been identified. This is the essential sign in the envelope argument. The package proves the root-order implication separately; it does not assume that every root lies on a decreasing branch.

[LeanSuffixAnalytic.typeThreeArea_lt_typeFourArea \(audit\)](#) [LeanSuffixAnalytic.typeThreeArea_descending_root_lt \(audit\)](#)

Let $\Delta(t) = P_3(r(t)) - P_4(t)$. Locally the roots are trapped in a compact subinterval of $(0, 1)$; continuity and uniqueness give continuity of r on $(0, 1]$. At an interior point the denominator $A_3'(r(t))$ is nonzero, so implicit differentiation and the variational identities yield

$$r'(t) = \frac{A_4'(t)}{A_3'(r(t))}, \quad \Delta'(t) = (r(t) - t)A_4'(t).$$

Thus Δ increases before t_* and decreases after t_* . At $t = 1$ the argument uses one-sided continuity, not a derivative of the singular radical there. Consequently

$$\Delta(t) \leq \Delta(t_*) = P_3(s_*) - P_4(t_*) < 0 \quad (0 < t \leq 1).$$

This produces an equal-area, strictly improving admissible type-(iii) competitor for every modeled four-arc curvature at this density.

[LeanSuffixAnalytic.equalAreaRoot_continuousOn_of_compact_unique \(audit\)](#) [LeanSuffixAnalytic.equalAreaRoot_hasDerivAt \(audit\)](#) [LeanSuffixAnalytic.equalAreaPerimeterGap_hasDerivAt \(audit\)](#) [LeanSuffixAnalytic.equalAreaPerimeterGap_le_fold \(audit\)](#) [LeanSuffixAnalytic.allCurvature_typeThreeImprovement_of_envelope \(audit\)](#) [CMVSuffixModel.FourArcCandidate.not_isWeightedPerimeterMinimizer_of_stationaryEqualAreaPair \(audit\)](#)

3. Certified density intervals

The remaining bounded-density task is exactly the six stationary-pair conditions in §2, with the stated branch inequalities. It is discharged by finite, exact-rational certificates, not floating-point samples. Interval operations have real-containment theorems; sine and cosine use Taylor polynomials of degrees 27 and 26 with proved remainders; inverse-trigonometric bounds include principal-branch checks. Centered mean-value enclosures retain the shared intermediate point in repeated derivative occurrences.

[LeanSuffixReflective.QInterval.RealContains \(audit\)](#) [ScalarSuffixCertificate.CertificateExpr.sound \(audit\)](#) [ScalarSuffixCertificate.sinTaylor27Interval_sound \(audit\)](#) [ScalarSuffixCertificate.cosTaylor26Interval_sound \(audit\)](#)

[ScalarSuffixCertificate.realContains_arcsin_of_sin_bounds](#) (audit)
[LeanSuffixAnalytic.centered_meanValue_exists](#) (audit)

For the lower part, the certificates use four angular coordinates x, y, w, v on explicitly checked subintervals of $(0, \pi/4)$, with $t_* = \cos x$ and $s_* = (1 + \cos w)/2$. The equations $C_4 = \cos x - \lambda \cos y = 0$ and $C_3 = \cos w - \lambda \cos v = 0$ encode incidence; the residuals $F = 0$ and $E = 0$ encode $K_4(t_*) = 0$ and equal area. The sign of G is the sign of the perimeter gap, and the checked K_3 expression has the sign of $A_3(s_*)$. The rational brick gives affine predictions and a closed four-dimensional error box. The continuous map $(-F, C_4, E, C_3)$ has each component negative on its lower face and positive on its upper face. Here the common-zero proof uses the triangular form of these particular equations: solve $C_4 = 0$ and $C_3 = 0$ by their continuous principal-angle projections in the second and fourth error coordinates, then apply the one-dimensional intermediate value theorem first to $-F$ in the first coordinate and then to E in the third. The face signs keep all selected coordinates in the closed box; the strict $K_3 < 0$ and $G < 0$ estimates therefore hold at the resulting common zero. The underlying one-dimensional lemma allows endpoint zeros and a degenerate interval; strict face signs are extra information supplied by these bricks. [ScalarSuffixCertificate.oppositeFace_zero](#) (audit) This explains why the certificates establish existence of a genuine stationary pair, not just small residuals near an approximate solution. [MiddleFaceAssembly.Brick](#) (audit)
[MiddleFaceAssembly.Model.rootMap](#) (audit) [MiddleFaceAssembly.Model.rootMap_continuous](#) (audit)
[MiddleFaceAssembly.CellCertificate](#) (audit) [MiddleFaceAssembly.CellCertificate.sound](#) (audit)
[MiddleFaceAssembly.Conditions.root_exists](#) (audit) [MiddleFaceAssembly.pointEquationSemantics](#) (audit) [MiddleFaceAssembly.Conditions.stationaryEqualAreaPair_exists](#) (audit)

The first prototype covers $[51/50, 102001/100000]$; the 113 middle-face cells cover $[102001/100000, 25781/25000]$; the right prototype covers $[25781/25000, 33/32]$. Their closed intervals meet at exact rational endpoints. The first prototype uses its explicit triangular root construction; the middle and right bricks use the reusable face assembly. Combined, they exclude every candidate for $51/50 \leq \lambda \leq 33/32$. [ScalarSuffixCertificate.FirstPrototypeCell.triangular_root](#) (audit)
[ScalarSuffixCertificate.FirstPrototypeCell.candidate_not_isWeightedPerimeterMinimizer](#) (audit)
[MiddleFaceCells.candidate_not_isWeightedPerimeterMinimizer](#) (audit)
[RightPrototypeCell.candidate_not_isWeightedPerimeterMinimizer](#) (audit)
[MiddleFaceCells.candidate_not_isWeightedPerimeterMinimizer_from_51_50](#) (audit)

On $[33/32, 9/7]$, 1024 compact boxes instead enclose s_*, t_* directly. Opposite signs of K_4 at the two t -faces locate its unique zero. Negative K_3 throughout the s -slab makes A_3 strictly decreasing there; opposite signs of $A_3 - A_4(t_*)$ at the two s -faces locate the equal-area root. A strictly negative enclosure of $P_3 - P_4$ completes the stationary comparison. These are the hypotheses of the generic compact-cell assembly theorem. [CompactCellAssembly.Conditions](#) (audit)
[CompactCellAssembly.Conditions.typeFourFold_existsUnique_in_slab](#) (audit)
[CompactCellAssembly.Conditions.typeThreeEqualArea_existsUnique_in_slab](#) (audit)
[CompactCellAssembly.Conditions.stationaryEqualArea_improvement_exists](#) (audit)
[CompactCellAssembly.Conditions.candidate_not_isWeightedPerimeterMinimizer](#) (audit)

The first two boxes are explicit nongenerated modules; the remaining boxes form the configured generated family. The final coverage theorem includes both explicit boxes and all generated boxes, so there is no omitted initial interval.
[CompactFirstCell.compactCandidate_not_isWeightedPerimeterMinimizer](#) (audit)
[CompactSecondCell.firstTwoBoxes_candidate_not_isWeightedPerimeterMinimizer](#) (audit)
[CompactCells1003To1023.bboxes0To1023_candidate_not_isWeightedPerimeterMinimizer](#) (audit)

4. The high-density competitor and the final splice

For a circular segment of unit chord and half-angle $\theta \in (0, \pi)$, set

$$a(\theta) = \frac{\theta - \sin\theta\cos\theta}{4\sin^2\theta}, \quad \ell(\theta) = \frac{\theta}{\sin\theta}.$$

The function a increases continuously from zero to infinity. Let $\theta(X)$ be its inverse and $L(X) = \ell(\theta(X))$. Then $L'(X) = 2\sin\theta(X)$. Define $g(X) = 2L(X) - L(2X)$ for $X > 0$. [area](#) (audit) [ell](#) (audit) [area_strictMonoOn](#) (audit) [area_surjOn](#) (audit) [thetaOf](#) (audit) [arc](#) (audit) [hasDerivAt_arc](#) (audit) [g](#) (audit)

The one-variable calculation locates the global minimum at $X_* = a(\theta_*)$, where $\theta_* \in (0, \pi/2)$ is the unique root of $3\theta - 3\sin\theta\cos\theta = \pi$. It gives $g(X) \geq g(X_*) = 3\cos\theta_*$. An exact rational trigonometric certificate then proves

$$\lambda \geq 1.2581840884, \quad X > 0 \quad \Rightarrow \quad \frac{1}{\lambda} < g(X).$$

The decimal in this display denotes the exact terminating rational, not a measured approximation. `F (audit)  $\theta_{\text{star}}$  (audit)  $g_{\theta_{\text{star}}}$  (audit)  $g_{x_{\text{star}}}$  (audit)  $\min_{g \in \text{three\_mul\_cos\_}\theta_{\text{star}}} (audit) \text{cmv\_comparison\_bound\_optimized} (audit)$`

For the candidate let X be the area of one exterior cap divided by c^2 . Replace the two congruent exterior caps by one upper cap of the same chord and twice their individual area, keeping the strip core and exposing its lower chord. The inverse $\theta(2X)$ constructs this cap on the full $(0, \pi)$ branch. Weighted area is unchanged, while the perimeter difference is exactly $P_{\lambda}(C) - P_{\lambda}(C_{\text{rep}}) = \lambda c \left(g(X) - \frac{1}{\lambda} \right) > 0$. The new lower chord is explicitly proved to lie in the frontier and contributes c , at density one. This is a geometric competitor with the canonical complete-frontier perimeter.

`FourArcCandidate.CapReplacement (audit) FourArcCandidate.capReplacementRegion (audit)`
`ReplacementAssembly.exposedLowerChord_subset_frontier (audit)`
`replacement_frontier_weightedPerimeter_eq (audit)`
`FourArcCandidate.cap_replacement_preserves_area (audit)`
`FourArcCandidate.cap_replacement_perimeter_difference (audit)`
`FourArcCandidate.cap_replacement_strictly_improves (audit) cmv_type_four_not_minimizing (audit)`

Finally, for $\lambda \geq 51/50$, use the lower certificates if $\lambda \leq 33/32$, the compact certificates if $33/32 < \lambda \leq 9/7$, and cap replacement otherwise. In the last case $1.2581840884 < 9/7 < \lambda$. Every case constructs an admissible equal-area strict improvement, contradicting the defining inequality for a minimizer. This is exactly the published root.

`CMVPublishedCutoff51_50.candidate_not_isWeightedPerimeterMinimizer_from_51_50 (audit)`

Scope and the remaining publication gap

The number 51/50 is the lower endpoint of this published certificate argument: its first certified brick begins there, the adjoining families cover upward to the cap-replacement range, and no certificate in this argument supplies the required stationary comparison below that endpoint. It is not asserted to be a sharp geometric transition. Extending this method toward density one requires rigorous control of the limiting geometry, descending equal-area branch, and strict comparison as the relevant margins become small; the present result supplies no conclusion for $1 < \lambda < 51/50$.

A publication of the full CMV conjecture would require substantially more than the exclusion proved here. The source regularity and candidate-classification arguments must be connected to the explicit coordinate constructors, including symmetry reductions, interface contact and refraction, limiting or degenerate configurations, and the precise source parameter conventions. The complete-frontier perimeter used here must be related, with the necessary representative and boundary-regularity hypotheses, to the source's measure-theoretic perimeter and its treatment of sets modulo area-null changes; arbitrary finite-perimeter sets cannot simply be identified with these carriers by name. Existence, compactness, regularity, and the exhaustive reduction of possible minimizers to the source candidate types also need their own justified source-to-model bridge. Finally the full conjectured classification and comparisons among the surviving types are not consequences of merely excluding the modeled type-(iv) family. The explicit carrier/frontier and scalar-realization theorems above resolve the comparisons used in this theorem, but do not replace those geometric, measure-theoretic, and source-to-model tasks.